Authenticated Personalized "Study with Me" Groups

A Platform for Social Engagement and Accountability

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ABSTRACT

The current form of social engagement in online programs and courses is discussion threads and forums. However, the various ways in which students connect have evolved to students studying on YouTube and Bilibili with streamers who live stream themselves studying for companionship, and viewers join and share their daily goals in the comments. Discord has seen a boom of educational servers where students show their aesthetic work desks or screen-share their work. All these students are from different time zones and different courses. No platform exists where students from the same class or college can study online together. This study shows the necessity for a large-scale social engagement platform authenticated with the university portal, removing the need for moderators or teaching assistants to monitor these students continuously. This study also analyses the different applications that currently exist to support this large-scale implementation. Quantitative research was conducted by sharing a survey containing a prototype of an authenticated study group with the human-computer interaction class from the Online Master's in Computer Science at Georgia Institute of Technology. The results showed that 84% of the students are willing to use this platform. Accountability tools have also been analyzed and no tool exists that guides students in real-time to improvise on their studying habits or guide them to approach things differently. With more educational programs available online, the future of learning at scale needs an authenticated platform for social engagement as well as personalized guidance.

CCS CONCEPTS

• Human-centered computing • Quantitative Research • Educational Technology • Online Education

KEYWORDS

Virtual Study Groups, Social Engagement, Accountability

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ACM Reference format:

FirstName Surname, FirstName Surname and FirstName Surname. 2024. Authenticated Personalized "Study With Me" Groups. In *Proceedings of the Tenth ACM Conference on Learning @ Scale (L@S '24), July 18-20, 2024, Atlanta, GA, USA. 4 pages.* https://doi.org/

1 Introduction

The "study with me" community [1] is increasing rapidly on YouTube and Bilibili, with more and more creators livestreaming themselves studying. The viewers tend to form a parasocial connection [2] with the content maker. The YouTubers use aesthetic work setups along with nature backgrounds and instrumental studying music. Aestheticism tends to make a student "romanticize" studying [3]. An all-time live YouTube video of the "Lofi girl" studying attracted a lot of users and inspired many YouTubers [4] by bringing a feeling of calmness and creating a comfortable virtual environment to study. It makes students feel connected to their past, which makes them feel like they belong and are part of a greater audience who is also currently studying with them. A study shows that aesthetic videos before the Covid-19 pandemic had words such as 'love' and 'beautiful' in their comments, but after the pandemic, the comments had words such as 'assignment' and 'study' [5].

Several Discord servers are educational servers with study groups moderated by admins. Study Together and StudyStream are one of the largest study group servers with over 400,000 and 200,000 users [6]. The common features in most of these groups are – accountability channels, muted video calls and screensharing, study tips channels, and updating session goals when you join the group. The accountability and study tips channels are where students share texts about their progress and goals along with helpful tools they use or studying habits they follow. The muted video calls and screensharing serve as a place where students can feel the social presence of another person who is also working hard and focusing on their studies. There are other channels too where students can unmute their mics and discuss what they have studied using the Feynman Studying Technique [7], work together listening to a shared playlist, as well as games or movies for a longer break. The whole platform serves as a mental model of a real-life classroom where students can see one

another, talk to one another during breaks, have a social life, and connect with each other.

2 Motivation

Wang, G. and Zhang, Y., in their study [8], observe12 famous streamers studying on Bilibili and preparing for an examination. Their main reason to live stream was to find company while studying for long hours at a stretch. The viewers are college students looking for company and motivation. The viewing effect of watching someone studying inspires another person to study [9]. Most of the streaming videos do not show the streamer completely, only their hands making notes or typing or programming. Even though no direct instructions are given on studying, students form a sense of community (SOC) through self-regulated learning (SRL). Most students on Discord do not know each other’s names. A study shows that anonymity proves to provide more social engagement [10].

Studies on predicting the drop-out rate of MOOCs [11] [12] show that social interaction helps students stay motivated and participate in discussions. In a shared study space, these discussions could take place more synchronously as one would find a classmate to discuss a topic with more quickly.

Productivity and accountability prove to be positive sources of motivation in a shared study space. The application Forest is used by more than 2 million users to sync timed study sessions, at the end of which the user plants a tree in their virtual garden [13]. It is used to follow the Pomodoro Studying Technique [14], and the Forest Study Chat server on Discord [15] has continuous updates on studying sessions starting every few minutes. Productivity also increases by viewing other students’ daily routines.

A negative outcome of a shared study group can be distractions. Students can get caught up in talking or viewing another student’s content. However, this distraction is for a short time, and eventually, students will see the other people studying and retain their focus.

3 Analysis of Existing Applications and Tools

Current applications that support study groups have no authentication with universities. For the same reason, students are from different time zones and age groups and study different subjects. Microsoft Teams is the only application that has a student account authenticated with the university portal. However, one must join the call to see who is currently studying, unlike Discord, where all the participants studying can be viewed. Discord does not provide complete authentication and has a complicated user-interface design [16], which does not allow it to be suitable for study groups. Other applications have been reviewed and analyzed in Figures 1 and 2. LifeAt Spaces [17] is the only platform that provides aesthetic virtual backgrounds; however, the application is based on subscription, and students need to pay for group studies.

The analysis of accountability tools in Figure 3 shows how different tools are used to track the number of hours studied or share to-do lists with friends; however, no real-time feedback or suggestions are provided to the user. To make the best use of this data, artificial intelligence can be used to interact with the user through a chatbot or a feature for daily tips and suggestions on improving their studying habits and approaching their work in different ways.

Product	Description	Pros	Cons
Discord	Video chat application	Set different channels, see who is studying on call without having to join the call, personalized usernames and avatars	No authentication, UI is complex
Microsoft Teams	Video chat application	Authentication, admin can control participants on call	No personalized usernames or avatars, cannot see who is studying without joining
Zoom	Video chat application	Admin can control participants on call	No authentication, chat history is not saved
Google Meet/Classroom	Video chat application	Admin can control participants on call	No authentication, chat history is not saved
Focusmate	Book a study session with a group on calendar	Calendar booking, personal profiles	No authentication, compulsory to keep video and mic on
Todoist	Update lists and chat with others	Productivity statistics	No video calling, no authentication

Figure 1: Analysis of Coworking Spaces – 1

Product	Description	Pros	Cons
LifeAt Spaces	Virtual dynamic backgrounds and study space	Aesthetic environment, mindfulness tool, to-do lists	Paid subscription for group study
My Focus Space	Book a study session with a group on calendar	Analysis for productivity, to-do lists	No authentication
Flow Club	Book a study session with a group on calendar	To-do lists	Paid Subscription, no authentication
Social Pomodoro	Randomly connect with someone to study with	Randomly connect with a user	No video calling, no authentication
Focused Space	Video chat application	Wake up calls, Community session, Accountability check-in	Paid subscription, no authentication
Study Together	Virtual background and study space	Analysis for productivity, to-do lists, aesthetic environment, mindfulness tool	No authentication, poor UI design

Figure 2: Analysis of Coworking Spaces – 2

Product	Description	Pros	Cons
Forest	Timer by planting trees with a group of people	Aesthetic timer, productivity statistics	No authentication
Marinara Timer	Directly starts pomodoro timers	Customization as well as automated timer	No authentication, no productivity statistics
Pomodor	Timer	Productivity Statistics	No authentication
FocusList		Calendar-wise productivity statistics	No authentication
Toggl Track	Track other people’s progress	Calendar-wise productivity statistics	No authentication
Flow, Be Focused	Timer	Calendar-wise productivity statistics	No authentication
Snaz	Timer – Countdown and Countup	Connect with OBS for live-streaming, add timer as a widget	No authentication
Fivable	Form a group and set timer, share to-do lists	Shared to-do lists, chat in-between	No authentication

Figure 3: Analysis of Accountability Tools

4 User-Interface Design

The proposed design is called Ed Study Groups, and the working prototype in Figma can be accessed [18]. It is designed as an extension to Ed Discussions [19]. The mobile application design of an authenticated study group consists of eight screens as shown in Figures 4 to 11.

The home page consists of the courses the student is enrolled in the current semesters as well as the previous semesters (Fig. 5). The card-based design was shared as a survey to students in the human-computer interaction class enrolled in the Online Master’s in Computer Science at Georgia Institute of Technology. The class uses Ed Discussion to connect with classmates as well as teaching assistants regarding the curriculum. Ed Discussion does not have a mobile application currently. The design is such that a student can use Ed Study Groups as a form of social engagement and use Ed Discussions for formal discussions. To switch from one to another, the account settings page (Fig. 11) has the option to open Ed Discussions.

A total of 73 participants participated in the survey. The

survey was shared with 6 participants not in the human-interaction course and who have not used Ed Discussion before. Overall, 84% of the students are willing to use an authenticated virtual study platform. The prototype is card-based, and Figure 12 shows that 93% of the participants, on average, found the eight screens of the mobile application easy to understand.

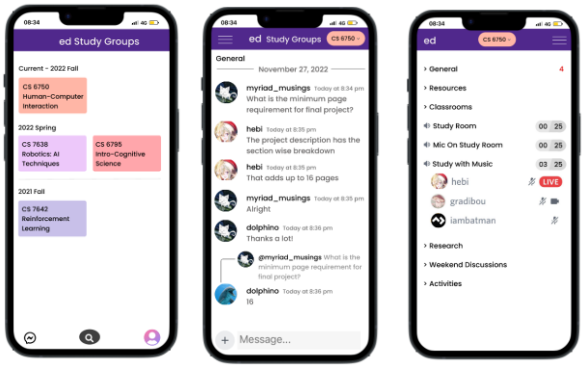

Figure 4: Home Page, Figure 5: CS 6750 study group chat, Figure 6: Navigation bar

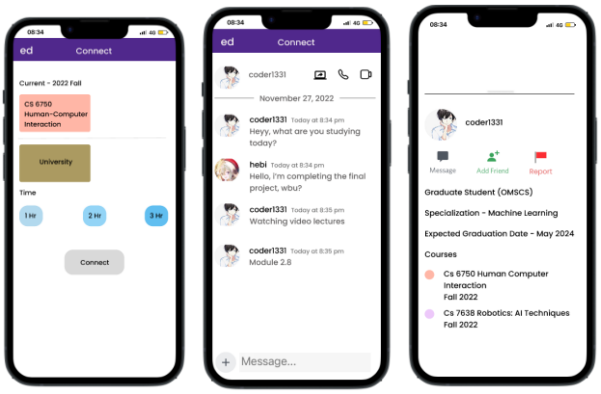

Figure 7: Connect feature, Figure 8: Connected with a friend chat screen, Figure 9: Profile Information

The connect feature (Fig. 7) shows how a student can connect with someone in the same course they are enrolled in currently or a student studying another subject in the university and is available to study for one, two, or three hours. The number of hours and course/university need to be selected before pressing the "connect" button. The feedback provided by one user was to not set the required hours beforehand but let the students decide the number of hours they want to study once they find a suitable friend.

Along with this connect feature, a group of students can listen to music together, have movie nights or game nights, hold weekend discussions, and teachers can conduct live office hours (Fig. 6). It is like a virtual classroom where there are different classes going on. One can switch from one class to another by selecting the drop-down menu of classes at the top of the screen or going to the home page and selecting the particular class (Fig. 5 and 6).

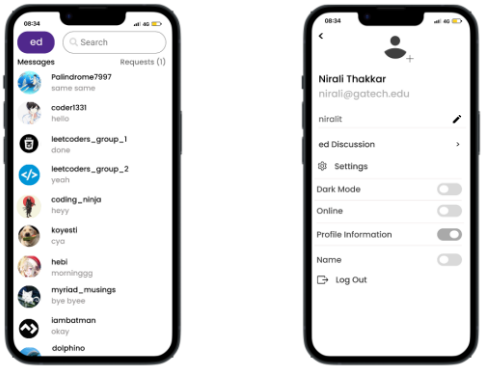

Figure 10: Messages, Figure 11: Account Settings

The profile information screen displays the courses a student has taken in the past semester along with their specialization and expected graduation date (Fig. 9). In this way, online students can also meet during the graduation ceremony. Figure 13 shows that users who did not use any form of timers before are willing to use productivity tools. Personalization mentioned in the survey was for usernames and avatars. 75% of the participants agreed to use customized names and avatars.

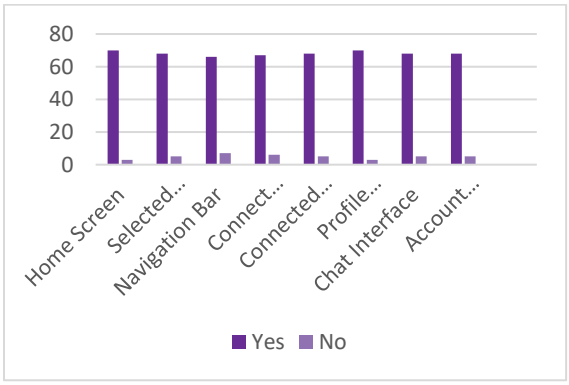

Figure 12: Eight screens of the UI design and its ease of understanding by the participants.

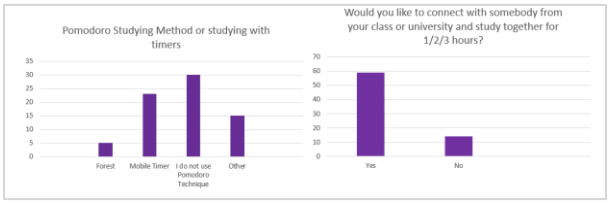

Figure 13: Participants who have not used timers or accountability tools are willing to do so after seeing the UI design.

5 Implementation

Figures 14-17 show the implemented web application.

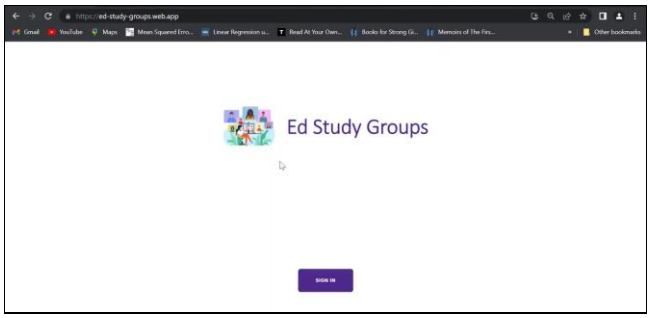

Figure 14: Implemented Web App Home Page

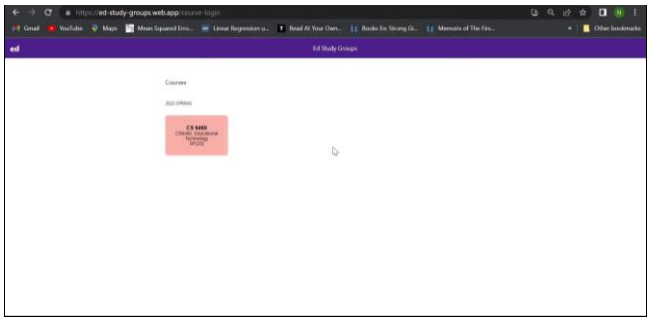

Figure 15: Implemented Web Course Home Page

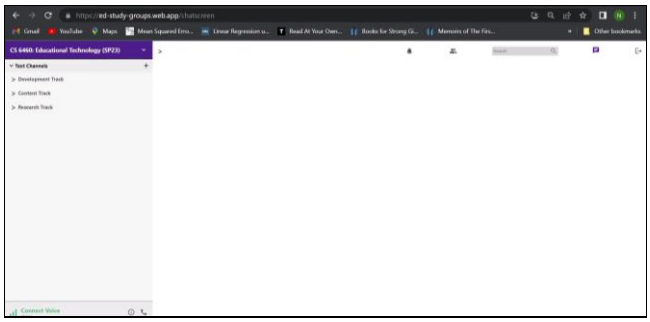

Figure 16: Implemented Web App Course Page for CS 6460

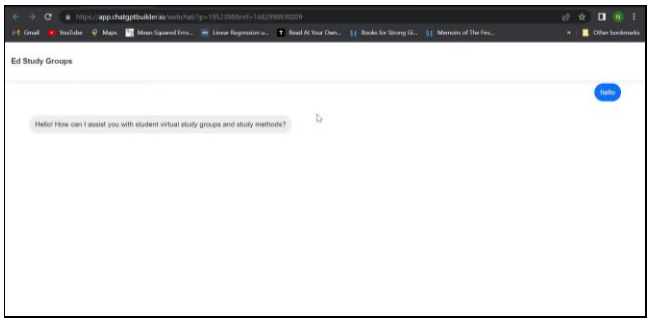

Figure 17: Implemented Web App Page for Accountability Chatbot

The current version authenticates users with their Google Sign-In Page instead of university portals. Students can create their own channels and conduct video calls. They can also connect with an accountability chatbot to receive advise on different studying techniques.

6 Implementation in Progress

(i) Personalized username settings need to be added in profile to show in video calls
(ii) A total of 35 accountability methods have been collected and the accountability tool needs to be trained on these methods.
(iii) The accountability tool needs to be connected with Canvas to access student grades and provide feedback.
(iv) And lastly, the web page needs authentication with university credentials.

7 Scaling

The scaling of an idea is equivalent to the additional cost per student [20]. The proposed idea of an authenticated virtual study group, along with personalized accountability tools, would require high maintenance for the privacy and protection of data as well as supporting thousands of video calls. An additional cost would be required for the architecture, but when divided by a large number of students, it would result in a minimal amount. For safety and privacy control, the warnings against a user for misconduct can be stored and reviewed instead of having teaching assistants supervise the group continuously. Furthermore, cost would be needed to maintain and provide all-time access to the personalized accountability tool.

8 Future Work

Future work includes applying machine learning and reinforcement learning to a user’s data to analyze their studying habits and automate their daily schedule and to-do lists. The software can learn the most productive hours of a student and suggest different ways of completing the most challenging task. It can learn when a student tends to procrastinate and suggest completing a small task like an assigned reading by converting it into an audiobook.

9 Conclusion

Students either form their own small groups or be a part of a community like “study with me” on YouTube [1] or Study Together and StudyStream Discord servers [6]. With more online degrees becoming available, social engagement platforms need to evolve from discussion threads to authenticated study groups that provide better connecting features and personalized settings. The proposed design of an authenticated group supports all these features. It need not be restricted to online programs, even on-campus students can solve their assignments together after classes on this shared platform. Students may not always join, the way they do not always participate in the forums, but they would always have a group of people with shared interests with whom they can study together.

ACKNOWLEDGMENTS

Thanks to Dr. David Joyner, Tyler Bobik, and Georgia Institute of Technology for allowing me to conduct the survey in the human-interaction class in the Online Master’s in Computer Science Program.

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